



INTEGRATED IT INFRASTRUCTURE | LIFE-SCIENCES MANUFACTURING

Engineering a Resilient Digital Backbone for Avantor, Panoli

D-TECH integrated compute, backup, cybersecurity, managed switching, structured cabling, rack and power into one enterprise-ready infrastructure platform.



CLIENT: AVANTOR PERFORMANCE MATERIALS INDIA LIMITED

PANOLI | ANKLESHWAR | GUJARAT

COMPUTE

2 Dell Servers

SWITCHING

5 Cisco Switches

CABLING

610 m Cat6

VALIDATION

24 Nodes

VERIFIED PURCHASE ORDER: 4900000941 | 9 OCTOBER 2017

Integrated infrastructure supply and installation | Commercial value omitted

A Complete IT Foundation - From Rack to Data Protection

The documented order combines 36 material and service lines into a single coordinated infrastructure deployment for the Panoli site.

VERIFIED TECHNOLOGY STACK

COMPUTE

2 x Dell R530

One server with HBA and one without HBA, creating the core enterprise compute layer.

BACKUP

LTO-6 + Backup Exec

Dell PowerVault external tape, Backup Exec server edition and Windows server agent.

MANAGED NETWORK

5 x Cisco SG300-28

Five 28-port managed gigabit switches with five Cisco support-service subscriptions.

STRUCTURED CABLING

40 Cat6 points

102 patch cords, two loaded 24-port panels, 40 jacks and faceplates, plus 610 m cable laying.

POWER & RACK

42U + 5 kVA UPS

Valrack 42U server rack, APC online UPS and 16 x 12V 18Ah batteries.

SECURITY & DLO

16 + 20 users

Endpoint protection for 16 users and DLO on-premise licensing for 20 users.

FULL-STACK DESIGN

Compute, network, backup, endpoint protection, physical infrastructure and power were procured as one coordinated programme.

42U Rack-Stack & Physical Infrastructure Design

A technical front-elevation concept mapping the documented rack, compute, switching, backup, power, cabling and protection materials.



Complete Documented Scope - Items 01 to 18

Every material and service line is listed with its original PO quantity. Commercial rates and values are intentionally excluded.

PHYSICAL FOUNDATION MATERIAL SCHEDULE

RACK • POWER • NETWORK • CABLING • SERVICES

SEE TECHNICAL SCHEMATIC — PAGE 3

01	AMP Tyco Cat6 SL Series Patch Cords	102 EA	10	Cisco SG300-28 28-Port Gigabit Managed Switch	5 EA
02	AMP Tyco Cat6 24-Port Loaded Jack Panel	2 EA	11	D-Link SC 100 Mbps Single-Mode Media Converter	4 EA
03	AMP Tyco Cat6 UTP Jack with Strain Relief	40 EA	12	PVC Casing and Capping - 25 mm	80 M
04	AMP Tyco 1-Port British-Style Shuttered Faceplate	40 EA	13	PVC Casing and Capping - 32 mm	40 M
05	Standard Back Box	2 EA	14	PVC Conduit Pipe - 32 mm	10 M
06	AMP Tyco Cat6 LSZH Cable Box - 1000 Feet	2 EA	15	Cisco CON-SNT Support for SG300-28 Switch	5 EA
07	AMP Tyco 12F SC Single-Mode Loaded FMS	1 EA	16	Gigabit NIC Card	8 EA
08	AMP Tyco SC-SC Duplex Single-Mode Fibre Patch Cord	2 EA	17	I/O Punching with Back Box and Faceplate	40 EA
09	Valrack 42U 800W x 1000D Server Rack	1 EA	18	Jack Panel Fitting Service	2 EA

Complete Documented Scope - Items 19 to 36

The second half covers implementation services, certification, cybersecurity, backup, servers and power protection.

FROM IMPLEMENTATION SERVICES TO DATA RESILIENCE

36/36 PO LINES DOCUMENTED

19	42U Rack Installation and Dressing Service	1 EA	28	Dell PowerVault LTO-6-200 External Tape Drive	1 EA
20	Managed Switch Installation Service	5 EA	29	APC SRC 5K Online UPS	1 EA
21	Casing and Capping Fitting Service	130 M	30	Exide 12V 18Ah Battery	16 EA
22	Cat6 Cable Laying Service	610 M	31	Open Battery Rack and Links	1 EA
23	Network Node Certification	24 EA	32	Symantec DLO On-Premise - 10 User Pack	2 EA
24	Symantec Endpoint Protection 14 - Per User	16 EA	33	Symantec Essential Support - 12 Months for DLO	2 EA
25	Symantec Backup Exec Agent for Windows Server	1 EA	34	Symantec Backup Exec Server Edition for Windows	1 EA
26	Symantec Essential Support - 12 Months	1 EA	35	Symantec Essential Support - 12 Months for Backup Exec	1 EA
27	Dell PowerEdge R530 Server with HBA Card	1 EA	36	Dell PowerEdge R530 Server without HBA Card	1 EA

ALL QUANTITIES VERIFIED AGAINST AVANTOR PURCHASE ORDER 4900000941

A Structured Path from Supply to Certified Connectivity

The order included installation, dressing, cable laying and node certification - moving beyond product supply to a complete site implementation.

01**RACK & POWER**

Install and dress the 42U rack; position the online UPS and battery system.

02**COMPUTE**

Deploy two PowerEdge R530 servers and connect the LTO-6 backup platform.

03**NETWORK**

Install five managed switches, fibre distribution and media conversion.

04**CABLING**

Lay Cat6 cabling, terminate outlets and organize patch panels and cords.

05**PROTECT**

Establish endpoint, DLO and Backup Exec software protection.

06**CERTIFY**

Test and certify 24 network nodes for documented readiness.

Delivery Assurance

**Single Accountability**

One partner coordinating products, installation services and infrastructure integration.

**12-Month PBG**

Balance payment linked to a performance bank guarantee valid for 12 months after installation completion.

**Site Delivery**

The documented delivery location and terms were aligned to Avantor's Panoli facility.

**Standards-Based Build**

LSZH Cat6, single-mode fibre, managed switching and node certification support maintainability.

IMPLEMENTATION SIGNATURE

A clean handoff from physical infrastructure to secure, backed-up and certified digital operations.

One Infrastructure Programme. Multiple Layers of Resilience.

The engagement demonstrates D-TECH's ability to deliver a complete enterprise IT foundation for a demanding life-sciences manufacturing environment.

2**ENTERPRISE SERVERS**

A dual-server compute foundation for business and data-protection workloads.

5**MANAGED SWITCHES**

A centrally manageable gigabit access layer with Cisco support coverage.

610m**CAT6 LAYING**

Structured copper cabling supported by fibre distribution and media conversion.

24**CERTIFIED NODES**

Tested network points providing documented implementation assurance.

What This Case Demonstrates

- ✓ **D-TECH can integrate infrastructure from multiple OEMs under one accountable delivery model.**
- ✓ **Server, backup and network design are stronger when implemented as one coordinated stack.**
- ✓ **Physical quality - rack dressing, LSZH cabling and certification - directly supports serviceability.**
- ✓ **Cybersecurity, user backup and enterprise tape protection create layered data resilience.**

DOCUMENTED ENGAGEMENT

Purchase Order: 4900000941, dated 9 October 2017

Client: Avantor Performance Materials India Limited

Delivery location: Panoli, Ankleshwar, Gujarat

Documented scope: 36 material and service lines | Commercial value: omitted for confidentiality